

Special Continuous Distribution:

$$\int_a^b f(x) dx = \frac{1}{b-a} \int_a^b dx$$

1. Continuous Uniform Distribution

$$f_X(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

$$= 0, \quad \text{otherwise}$$

$$E(X) = \frac{a+b}{2}, \quad V(X) = \frac{(b-a)^2}{12} = \sigma^2$$



$$M_X(t) = E(e^{tx}) = \int_a^b e^{tx} \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{e^{tx}}{t} \right]_a^b$$

$$M_X(t) = \frac{e^{tb} - e^{ta}}{t(b-a)}, \quad t \neq 0$$

$$= 1, \quad t = 0$$

2. Exponential Distribution:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$= 0, \quad \text{otherwise}$$

$$M_X(t) = \frac{\lambda}{\lambda - t}, \quad t < \lambda$$

$$E(X) = \left(\frac{1}{\lambda} \right) = (2) \Rightarrow \lambda = \frac{1}{2} \quad \therefore f(x) = \frac{1}{2} e^{-\frac{x}{2}}, \quad x > 0$$

$$V(X) = \frac{1}{\lambda^2}$$

$$f(x) = 2e^{-2x}, \quad x > 0$$

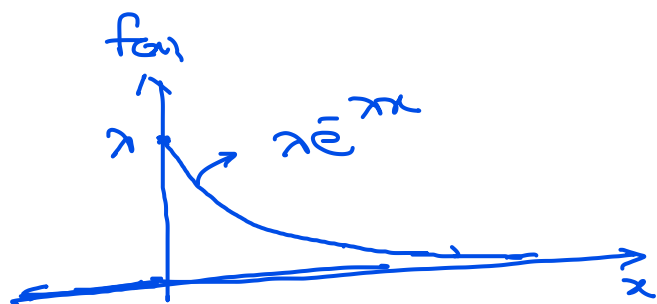
$$F(x) = \begin{cases} 0, & x < 0 \\ 1 - e^{-\lambda x}, & x \geq 0. \end{cases}$$

$$F(x) = P(X \leq x)$$

$$\underline{P(X > t)} = 1 - P(X \leq t)$$

$$= 1 - F(t)$$

$$= 1 - (1 - e^{-\lambda t}) = \underline{e^{-\lambda t}}$$



✓ Memoryless property: $X \sim \text{Exp}(\lambda)$, then

$$\underline{P(X > t+h | X > t)} = \underline{P(X > h)} \text{ for any } t > 0, h > 0.$$

Proof:- $P(\underbrace{X > t+h}_A | \underbrace{X > t}_B) = \frac{P(A \cap B)}{P(B)}$

$$= \frac{P(X > t+h, X > t)}{P(X > t)}$$



$$= \frac{P(X > t+h)}{P(X > t)}$$

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$$= \frac{e^{-\lambda(t+h)}}{e^{-\lambda t}} = e^{-\lambda h} = P(X > h)$$

Shifted Exponential Distribution: $X \sim \text{Exp}(\mu, \sigma)$

✓ $f_X(x) = \frac{1}{\sigma} e^{-\frac{x-\mu}{\sigma}}, x > \mu, \sigma > 0, -\infty < \mu < \infty$

$$E(X-\mu)^k = \int_{\mu}^{\infty} (x-\mu)^k \cdot \frac{1}{\sigma} e^{-\frac{x-\mu}{\sigma}} dx$$

$$f(x) = \left(\frac{1}{\sigma}\right) e^{-\left(\frac{x}{\sigma}\right)}, x > 0$$

$$= \sigma^k \left(\int_0^{\infty} y^k e^{-y} dy \right)$$

$$\frac{x-\mu}{\sigma} = y$$

$$\Rightarrow \frac{1}{\sigma} dx = dy$$

$$= \sigma^k \int_0^{\infty} y^{(k+1)-1} e^{-y} dy$$

$$= \sigma^k \Gamma(k+1)$$

$$= k! \sigma^k$$

$$E(X - \mu) = \sigma \Rightarrow E(X) - \mu = \sigma$$

$$\Rightarrow \boxed{E(X) = \mu + \sigma}$$

$$E(X - \mu)^2 = 2\sigma^2 \Rightarrow E(X^2 - 2\mu X + \mu^2) = 2\sigma^2$$

$$\Rightarrow E(X^2) - 2\mu E(X) + \mu^2 = 2\sigma^2$$

$$\Rightarrow E(X^2) - 2\mu(\mu + \sigma) + \mu^2 = 2\sigma^2$$

$$\Rightarrow E(X^2) = 2\sigma^2 + 2\mu\sigma + \mu^2$$

$$V(X) = E(X^2) - (E(X))^2$$

$$= 2\sigma^2 + 2\mu\sigma + \mu^2 - (\mu + \sigma)^2$$

$$\text{or } \boxed{V(X) = \sigma^2}$$

$$M_X(u) = E(e^{uX})$$

$$= \int e^{ux} f(x) dx$$

$$= \int_{-\infty}^{\infty} e^{ux} \cdot \frac{1}{\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \int_{-\infty}^{\infty} e^{u(\mu + \sigma y)} e^{-y^2} dy$$

$$= e^{\mu u} \int_{-\infty}^{\infty} e^{y(\sigma u - 1)} dy$$

$$= e^{\mu u} \frac{e^{y(\sigma u - 1)}}{\sigma u - 1} \Big|_{-\infty}^{\infty},$$

$$\frac{x - \mu}{\sigma} = y$$

$$\Rightarrow \underline{x = \mu + \sigma y}$$

$$= \frac{e^{\mu u}}{\sigma u - 1}, \quad \text{provided } \sigma u < 1$$

$$M_X(u) = \frac{e^{\mu u}}{1 - \sigma u}, \quad \sigma u < 1$$

$u < \frac{1}{\sigma}$

$$\sigma u - 1 < 0$$

$$e^{\gamma(\sigma u - 1)}$$

$$\rightarrow 0 \text{ as } \gamma \rightarrow \infty$$

Ex: The time to failure in months, X of the 'light' bulbs produced at two manufacturing plants A and B obeys exponential distribution with means 5 and 2 months respectively. Plant B produces three times as many bulbs as plant A. The bulbs are mixed and sold. What is the probability that a randomly selected bulb will burn at least 5 months?

Soln: Let X denote life of bulb

$$X|A \sim f(x) = \frac{1}{5} e^{-\frac{x}{5}}, \quad x \geq 0$$

$$X|B \sim f(x) = \frac{1}{2} e^{-\frac{x}{2}}, \quad x \geq 0$$

$$P(A) = \frac{1}{4}$$

$$P(B) = \frac{3}{4}$$

$$P(X > 5|B) = 1 - P(X \leq 5)$$

$$= 1 - (1 - e^{-\frac{5}{2}})$$

$$= e^{-\frac{5}{2}}$$

$$P(X > 5) = P(X > 5|A)P(A) + P(X > 5|B)P(B)$$

$$= e^{-1} \left(\frac{1}{4} \right) + e^{-\frac{5}{2}} \left(\frac{3}{4} \right)$$

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